KELLY WANG SZE QING

HP: +6594687813 | Email: kellywang@u.nus.edu | github.com/kellywsq03 | linkedin.com/in/kelly-wang-sq/
Availability: Sep 2026 – Jun 2027 (Intern), Jun 2027 (Full-time)

EDUCATION

National University of Singapore (NUS) **Aug 2023 – Jun 2027**
Bachelor of Computing in Computer Science (GPA 4.81/5.00)

- **Awards**: Dean's List (GPA 4.94), ASEAN Undergraduate Merit Scholarship

WORK EXPERIENCE

Full-Stack Developer Intern, Singapore Power Group **Jan 2026 - Jun 2026**

- Engineered a scalable IoT sensor data ingestion pipeline supporting a planned nationwide rollout to 12,000 Singapore substations by implementing backend microservices (Go) with event streaming (Apache Kafka) and comprehensive unit testing
- Led the end-to-end solutioning and implementation of a full-stack (React, Django) AI chatbot, enabling power grid operators to query power grid telemetry (MongoDB) and user documentation (Vector RAG) through natural language.
- Achieved production-grade deployment (Docker, Kubernetes), monitoring with LLM observability tools (Langfuse) and automated CI/CD (Jenkins), positioning it for adoption by Singapore’s Grid Digital Twin.
- Improved the accuracy of chatbot data retrieval and analysis by engineering an agentic workflow that dynamically orchestrates deterministic tools for data retrieval (MongoDB), computation, and chart generation (eCharts).
- Spearheaded software development by collaborating with 10 senior engineers on code reviews and architectural decisions, accelerating development through agentic AI workflows, and aligning project requirements to business needs with project managers. Delivered a live product demo to 20 senior engineers.

Teaching Assistant, Programming Methodology II, National University of Singapore **Aug 2024 – May 2025**

- Strengthened programming proficiency among 25 CS undergraduates by delivering 10 weeks of programming labs on modern object-oriented and functional paradigms in Java and statically typed programming languages.
- Nominated for teaching awards by 2 students and received positive feedback for clarity of instruction.

PROJECTS

B.Comp. Dissertation: Extending java-slang for Source Academy **Aug 2026 – Apr 2027**

- Extending Source Academy's (NUS School of Computing) open-source Java interpreter (TypeScript), used by ~800 NUS Computing students annually to write and run Java in-browser, by implementing language features within an AST interpreter.
- Developing an educational visualization tool to enhance understanding of static typing, compile-time and runtime type behavior, type erasure, wildcards, and generics.

ClockedIt: Multi-Agent Coordination Middleware **Aug 2026**

- Architected and built a multi-agent coordination middleware, supporting an agent-router protocol robust against 30+ server tests, resolving cross-file stale-read conflicts currently unaddressed by file-locking and Git.
- Integrated the middleware with Codex agent harnesses, runtimes, and event streaming (TypeScript/Fastify).
- Led a 5-person team in defining layered architecture and cross-layer contracts for parallel development.

Simfella: Collaborative Web-Based Systems Dynamics Modelling Tool **Jan 2026 – Apr 2026**

- Developed and deployed a real-time (WebSockets) collaborative web app (React, Go) for time-stepped systems dynamics modeling and simulations, piloted for 600 RC4 students, which outperformed the incumbent educational tool by 3 points on a 6-point scale as part of a university research initiative on improving usability of education tools.

PlaneFella: Travel Itinerary Platform and Web Scraper **Jan 2026**

- Established an AI-powered travel planning full-stack web app (React, Flask) by integrating web scraping (Playwright), video transcription (Whisper), and itinerary generation (Gemini) in a 3-person team.

RespondR: Apple VisionPro AR/VR Fire-Response Training **Aug 2026**

- Engineered a Vision Pro and companion iOS app (Swift, RealityKit) that fuses LiDAR room-scanning with real-time 3D fire-outbreak simulation, delivering an immersive AR/VR firefighter training tool at Spatial Hack AI.

Roomie: Social Media App and Mobile Game **May 2024 – Jul 2024**

- Developed a touch-screen isometric mobile game (Unity, C#, PlayFab) featuring a grid-based furniture placement system in a 2-person team in 180 self-directed hours with no formal instruction as part of NUS’s independent software development program. Independently learned to model 3D assets (Blender)

SKILLS

- **Languages**: Python, JavaScript, Typescript, Go, Java, C++, Swift, SQL, R, HTML/CSS
- **Frameworks**: React, Node.js, Flask, Django, FastAPI, JavaFX, Junit, Unity, GraphQL
- **AI/ML:** LangChain, LangGraph, LangFuse, TensorFlow, PyTorch, Numpy, RAG, LLM Integration
- **Cloud & DevOps**: Docker, Kubernetes, Jenkins, CI/CD, Apache Kafka, AWS, Microservices
- **Others**: MongoDB, PostgreSQL, Redis, Full-Stack Development, Unit Testing, Git, AI-Assisted development